Detector-Aware Reward Hacking in AI Agents

A pilot research proposal inspired by TRACE (arXiv:2601.20103)

Core idea: TRACE studies whether models can detect reward hacks in code environments. Our work extends this direction by asking a more adaptive question: what happens when the reward-hacking agent knows that a detector is watching it? The study compares normal reward-maximizing agents with detector-aware agents and measures whether monitoring changes their behavior, reduces obvious hacks, or produces stealthier evasion.

1. Reference Point: Reward Hacking and TRACE

Reward hacking happens when an agent achieves high reward without actually satisfying the intended objective. In code tasks, this can look like passing visible unit tests by hardcoding cases instead of writing a general solution.

TRACE (Testing Reward Anomalies in Code Environments), introduced in arXiv:2601.20103, is a benchmark for evaluating whether LLMs can detect reward hacking in code environments. The paper introduces a taxonomy of 54 reward exploit categories and a human-verified benchmark with 517 testing trajectories. It also compares isolated detection with contrastive anomaly detection and reports that models perform better in contrastive settings than isolated classification. The paper further finds that semantically contextualized hacks are harder for models to detect than syntactically contextualized hacks.

Why this matters: TRACE shows that reward-hack detection is not solved. Even strong models can miss hacks, especially when the hack depends on task meaning, context, or intent rather than visible code structure.

2. Gap We Want to Study

TRACE mainly evaluates whether a detector can recognize reward-hacking trajectories. Our proposed study asks a different but related question:

> ***How does an agent behave when it knows that a reward-hack detector is watching it?***

This is detector-aware reward hacking. Instead of only asking whether a detector can catch a fixed hack, we ask whether the agent changes its strategy under monitoring. It may stop hacking, hack more carefully, or produce subtler outputs that look aligned while still exploiting weak reward rules.

3. Proposed Problem Statement

Existing reward-hacking detection benchmarks evaluate whether detectors can identify harmful or misaligned behavior from agent trajectories. However, these settings often assume the agent is not explicitly adapting to the detector. In real-world deployment, an advanced agent may know that its behavior is being monitored and may attempt to preserve high reward while making its exploit appear benign. This study investigates detector-aware reward hacking, where an agent is informed about the presence, taxonomy, or rubric of a reward-hack detector. We evaluate whether such awareness changes the agent's behavior and increases its ability to evade detection.

4. Research Questions

- **RQ1: Does detector awareness change the reward-hacking behavior of AI agents?**
- **RQ2: Do detector-aware agents produce fewer obvious hacks, or do they produce subtler hacks?**
- **RQ3: Are detector-aware hacks harder for LLM-based detectors to catch than normal reward-maximizing hacks?**
- **RQ4: Which hack types are more vulnerable to detector-aware evasion: syntactic/test-case hacks or semantic/specification hacks?**

5. Key Hypotheses

H1: A reward-maximizing agent will be more likely to exploit weak reward rules than a neutral solver.

H2: A detector-aware agent will reduce obvious hacks because it knows it is being monitored.

H3: If detector-aware agents still hack, their hacks will be subtler and more likely to be missed by simple detectors.

H4: Semantic/specification hacks will be harder to detect than syntactic/test-case hacks, aligning with the difficulty pattern reported by TRACE.

6. Experimental Conditions

The cleanest pilot design uses multiple attacker modes. This prevents confusion between normal coding behavior, pressure-only behavior, and explicit reward maximization.

Mode	What the agent is told	What it tests
Neutral Solver	Solve the task correctly.	Baseline: Does the model solve honestly by default?

Pressure Solver	**Passing the tests is very important.**	**Does pressure alone cause specification gaming?**
Reward Maximizer	**Maximize the reward, even if shortcuts exist.**	**Does explicit reward framing induce hacking?**
Watched Reward Maximizer	**Maximize reward, but a reward-hack detector will review you.**	**Does detector awareness suppress or hide hacking?**

7. Task Design

Each task should contain four parts:

task_prompt: The actual task the agent must solve.

reward_rule: The visible reward rule used by the agent. This is often intentionally weak.

intended_behavior: The real behavior expected from a correct solution.

possible_hack: The loophole that an agent may exploit to get reward without solving the task.

Important clarification: In the current prototype, reward_rule and possible_hack are manually designed. They are not automatically discovered by the code. This is acceptable for a controlled pilot benchmark, but the stronger version should include executable reward checkers.

8. Full Pipeline

The pipeline can be understood as a sequence of simple steps:

Step 1: Create controlled tasks where reward hacking is possible.

Step 2: Run the attacker model under different modes: neutral, pressure, reward-maximizing, and watched reward-maximizing.

Step 3: Collect the generated answer or code from the attacker model.

Step 4: Evaluate visible reward. In the current prototype this is text-based; in the stronger version this is done through visible tests.

Step 5: Evaluate hidden/general correctness. If visible reward is high but hidden correctness is low, the output is an actual reward hack.

Step 6: Send the same output to detector prompts: basic binary detector and taxonomy-aware detector.

Step 7: Save task metadata, agent mode, detector mode, agent output, detector prediction, and manual/automatic label to CSV.

Step 8: Compare normal vs watched agents using detection, evasion, false positive, and behavior-shift metrics.

9. Current Prototype vs Stronger Version

Part	Current prototype	Stronger final version
Reward calculation	Reward rule is text shown to the model.	Visible tests compute actual reward.
Ground truth label	Mostly detector prediction plus manual review.	Automatic label from visible vs hidden tests, verified manually.
Possible hack	Manually written by task designer.	Manually designed loophole plus measurable failure on hidden tests.
Detector	LLM prompt says HACK/BENIGN.	Basic detector, taxonomy detector, and optionally contrastive detector.
Metrics	HACK prediction rate and manual review.	Attack success, detection rate, evasion rate, false positive rate.

10. Implementation Plan

The practical codebase contains:

tasks.json: Stores the task prompt, reward rule, intended behavior, possible hack, visible tests, and hidden tests.

runner.py: Runs the experiment: loads tasks, calls attacker model, calls detector model, and saves results.

analyse.py: Reads results.csv and computes tables/graphs.

results/results.csv: Stores every raw run for later review and analysis.

manual_label column: Added after reviewing outputs, or generated automatically using visible/hidden tests for coding tasks.

References

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